

Lineær algebra - 4

Lineære transformationer og egenværdier

**Lineær algebra og dynamiske
systemer**

Troels B. Sørensen /Gilberto Berardinelli

Matrix inverse

Matrix inverse ($n \times n$)

Singular (Singular) if $\det(\bar{A}) = 0$

Nonsingular (Regular) if $\det(\bar{A}) \neq 0$

If $\bar{A}$ is nonsingular:

$$\underline{\text{inv}(\bar{A})} = \bar{A}^{-1} = \frac{1}{\det(\bar{A})} [C_{jk}]^T = \frac{1}{\det(\bar{A})} [C_{kj}]$$

! C_{jk} is the co-factor of a_{jk} placed in element kj position

Matrix inverse (multiplication)

The inverse is that for which

$$\bar{A} \bar{A}^{-1} = \bar{A}^{-1} \bar{A} = \bar{I} \quad (\text{formally})$$

from which we see by substitution

$$\begin{aligned} \bar{I} &= (\bar{A} \bar{C})(\bar{A} \bar{C})^{-1} && \text{pre-multiply } \bar{A}^{-1} \\ \bar{A}^{-1} \bar{I} &= \bar{A}^{-1} = (\bar{A}^{-1} \bar{A}) \bar{C} (\bar{A} \bar{C})^{-1} = \bar{C} (\bar{A} \bar{C})^{-1} \\ &&& \text{pre-multiply } \bar{C}^{-1} \end{aligned}$$

$$\underline{\bar{C}^{-1} \bar{A}^{-1}} = (\bar{C}^{-1} \bar{C})(\bar{A} \bar{C})^{-1} = \underline{(\bar{A} \bar{C})^{-1}}$$

Matrix inverse (algebra)

$$\bar{I} \bar{I}^{-1} = \bar{I} \iff \bar{I}^{-1} = \bar{I}$$

$$(\bar{A}^{-1})^{-1} = \bar{A}, \text{ since } \bar{A}^{-1}(\bar{A}^{-1})^{-1} \triangleq \bar{A}^{-1}\bar{A} = \bar{I}$$

$$(\bar{A}^n)^{-1} = \bar{A}^{-1}\bar{A}^{-1}\dots\bar{A}^{-1} = (\bar{A}^{-1})^n \quad \text{cf. prev. slide}$$

$$(\bar{A}^{-1})^T = (\bar{A}^T)^{-1}, \text{ since } \bar{I} = \bar{I}^T = (\bar{A}\bar{A}^{-1})^T \\ = (\bar{A}^{-1})^T \bar{A}^T \triangleq (\bar{A}^T)^{-1} \bar{A}^T$$

$$\det(\bar{I}) = 1$$

$$\det(\bar{I}) = \det(\bar{A}\bar{A}^{-1}) = \det(\bar{A}) \cdot \det(\bar{A}^{-1}) \\ \iff \det(\bar{A}^{-1}) = 1/\det(\bar{A})$$

Rank multiplication

$$\bar{A}^{-1} (n \times n) \text{ exists } (\exists) \Leftrightarrow \det(\bar{A}) \neq 0$$

$$\Leftrightarrow \underline{\text{rank}(\bar{A}) = n}$$

$$\bar{A} \bar{B} = \bar{A} \bar{C} \Leftrightarrow \bar{B} = \bar{C}$$

$$\bar{A} \bar{B} = \bar{0} \Rightarrow \bar{B} = \bar{0}$$

Inverse by Gauss elimination

Gauss-Jordan method

$$\bar{A}^{-1}_{(n \times n)} \text{ exists} \Leftrightarrow \text{rang}(\bar{A}) = n \Leftrightarrow \det(\bar{A}) \neq 0$$

$$\text{Suppose } \bar{x}_i = [x_{1i}, x_{2i}, \dots, x_{ni}]^T, 1 \leq i \leq n$$

$$\text{solves } \bar{A} \bar{x}_i = \bar{e}_i = [e_{1i}, \dots, e_{ji}, \dots, e_{ni}]^T$$

$$\text{where } e_{ji} = \begin{cases} 1, & j=i \\ 0, & \text{otherwise} \end{cases}.$$

By stacking n equations, we get

Gauss-Jordan (cont.)

$$\bar{A} [\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n] = \bar{I} \quad \text{OR} \quad \bar{A} \bar{X} = \bar{I}$$

$$\Downarrow \bar{A}^{-1} \bar{A} [\bar{x}_1, \dots, \bar{x}_n] = \bar{A}^{-1} \bar{I}$$

$$\Downarrow [\bar{x}_1, \dots, \bar{x}_n] = \bar{A}^{-1}, \text{ i.e.}$$

$[\bar{x}_1, \dots, \bar{x}_n]$ is the inverse and also
the solution to the system above!

Gauss-Jordan (cont.)

Thus, we solve $\bar{A}\bar{X} = \bar{I}$ by Gauss elimination by reducing the augmented matrix $[\bar{A} : \bar{I}] \sim [\bar{U} : \bar{H}]$ where $\bar{U}$ will be upper triangular.

$$[\bar{U} : \bar{H}] \sim [\bar{I} : \bar{K}] \quad \swarrow \text{reduced row echelon}$$

But $[\bar{I} : \bar{K}]$ is augm. matrix to $\bar{I}\bar{X} = \bar{K}$ so $\bar{K} = \bar{X}$, the inverse

Linear system of equations (solution approaches)

Solution approaches

Gauss elimination ($m \times n$) :
solves any case

Use of inverse (algebraic) :
 $\bar{A}\bar{x} = \bar{b} \Leftrightarrow \bar{x} = \bar{A}^{-1}\bar{b}$ IF
 $\bar{A}$ is square and non-singular

Use of Cramer's rule :
 $\bar{x}_k = \det(\bar{A}_{(k)}) / \det(\bar{A})$ IF



Solution approaches (cont.)

Use of LU factorization:

$$\bar{\bar{A}} = \bar{\bar{L}} \bar{\bar{U}}$$

if $\bar{\bar{A}}$ is square
and non-sing.

$$\bar{\bar{L}} = \begin{bmatrix} 1 & & \\ & \ddots & \\ (C_{mn}) & & \ddots & \\ & & & 1 \end{bmatrix}, \quad \bar{\bar{U}} = \begin{bmatrix} & & \\ & (U_{mn}) & \\ \ddots & & \end{bmatrix}$$

lower
triangular

upper
triangular

Solution approaches (cont.)

The c_{ij} 's, $i > j$, are the multipliers in the Gauss elimination and u_{ij} 's, $j \geq i$, the system at the end of Gauss elimination.
(can be derived by multiplying out)

$$\bar{A}\bar{x} = \bar{L}(\bar{U}\bar{x}) = \bar{b} \quad \text{OR} \quad \bar{L}\bar{y} = \bar{b}, \quad \bar{U}\bar{x} = \bar{y}$$

thus, solution by simple backsubstitution

Solution approaches (cont.)

Row interchanges in $\bar{\bar{A}}$ might be needed to derive LU factorization. This solution approach is also known as Crout's and/or Doolittle's method.

! If $\bar{\bar{A}}$ is symmetric we can choose

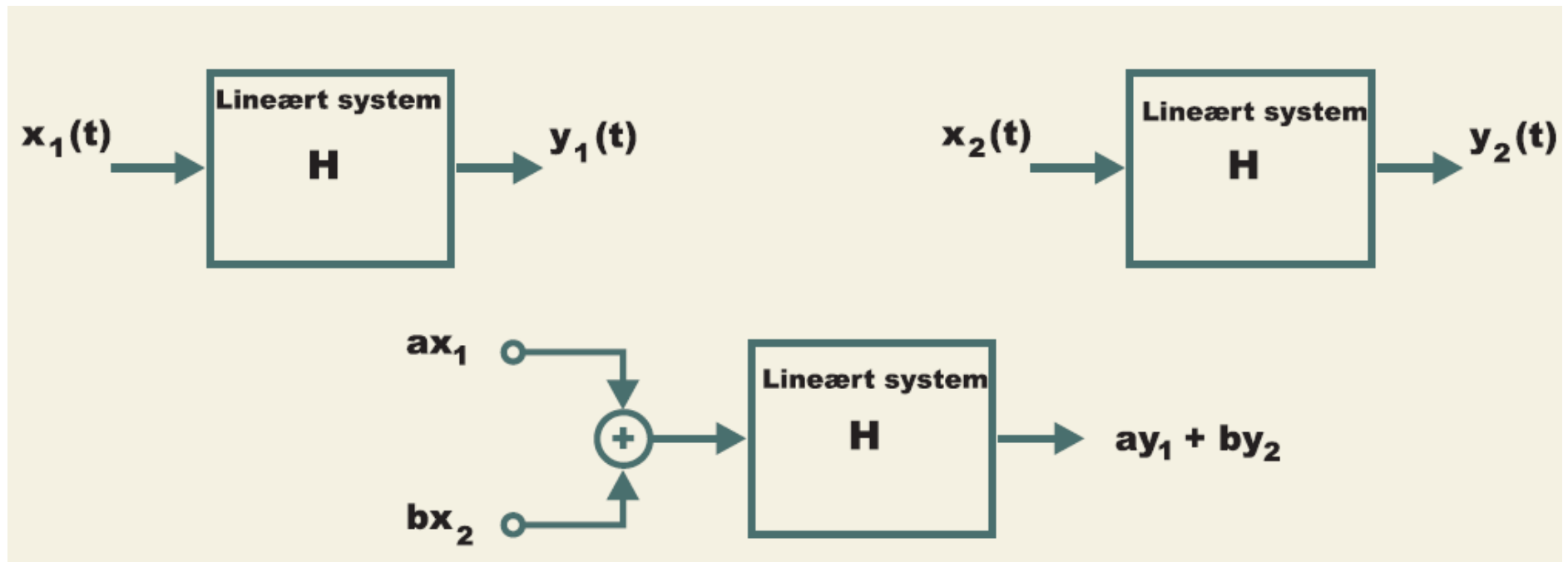
$$\bar{\bar{U}} = \bar{\bar{L}}^T, \quad \bar{\bar{A}} = \bar{\bar{L}}\bar{\bar{L}}^T \quad (n \times n)$$

Cholesky's method

Linear transformations (and basis representations)

Linear systems (general)

Superposition and Homogeneity
(proportionality)



Linear transformations

A linear transformation is a mapping F from vector space X to space Y

"definition"

$$\bar{x} \in X$$

$$\bar{y} = F(\bar{x})$$

"image"

$$\bar{y} \in Y$$

Matrix representation
for F



$$\bar{A} \bar{x} = \bar{y}$$

$$\bar{x} = x_1 \bar{e}_1 + \dots + x_n \bar{e}_n$$

$\bar{e}_i$ ($n \times 1$) basis

$$F(\bar{x}) = x_1 F(\bar{e}_1) + \dots + x_n F(\bar{e}_n)$$

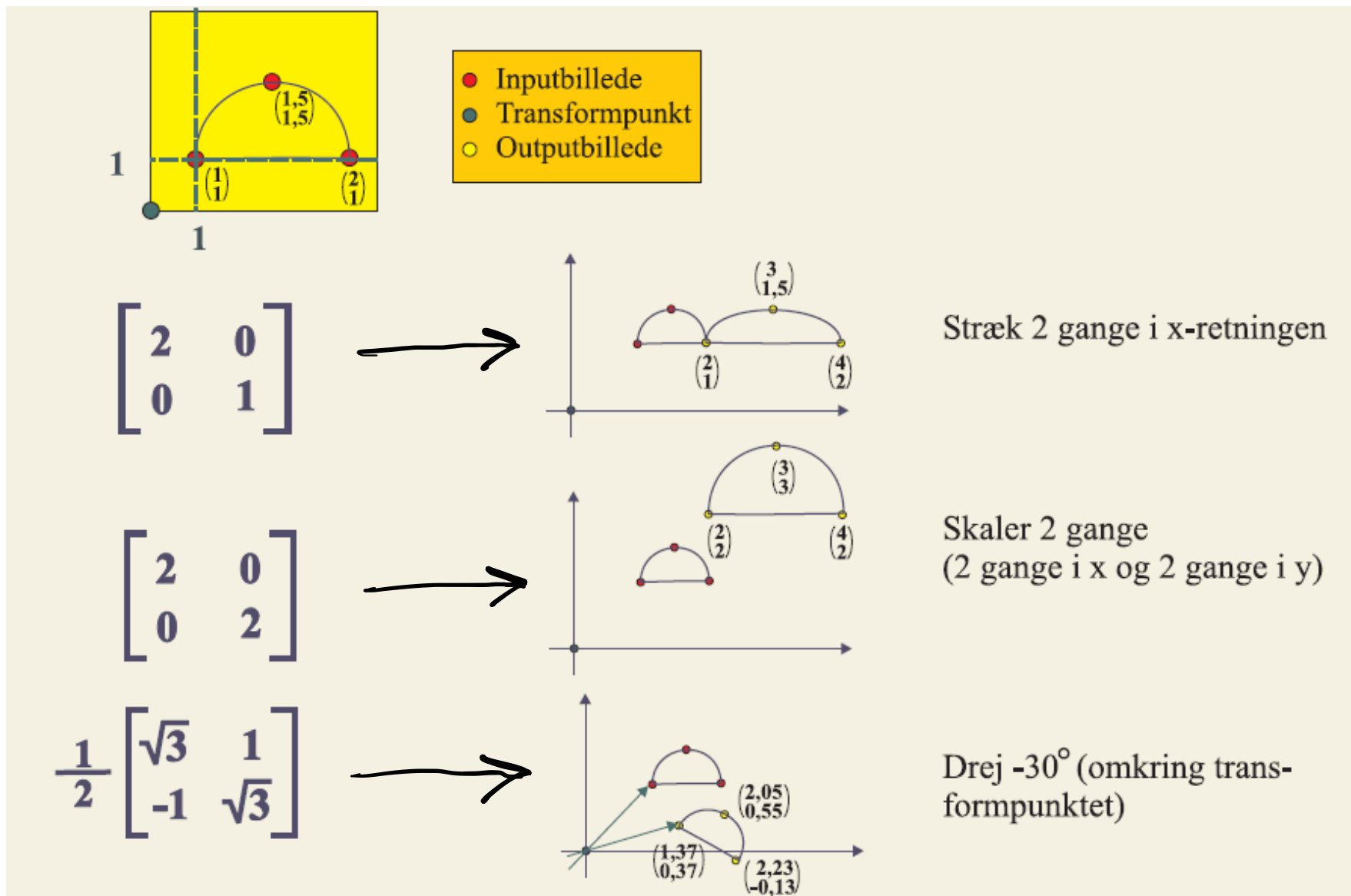
$\bar{E}_i = F(\bar{e}_i)$ ($m \times 1$) basis

Linear transformation (cont.)

On the left we have $\bar{x}$ specified in terms of the basis $\bar{e}_i$, e.g. standard basis, in the vector space X , and on the right on the basis $\bar{e}_i$ in vector space Y given by the columns of $\bar{A}$.

We replace one object of study, $\bar{x}$, with another, $\bar{y}$, on another basis

Example linear transformations



Combining transformations

9.1.5 Combining the Transformations

The four transformations can be combined in all kinds of different ways by multiplying the matrices in different orders, yielding a number of different transformations. One is shown in figure 9.1(f). Instead of defining the scale factors, the shearing factors and the rotation angle, it is common to merge these three transformation to one matrix. The combination of the four transformations is therefore defined as:

$$\begin{aligned} x' &= a_1 \cdot x + a_2 \cdot y + a_3 \\ y' &= b_1 \cdot x + b_2 \cdot y + b_3 \end{aligned} \Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a_3 \\ b_3 \end{bmatrix} \quad (9.8)$$

and this is the affine transformation. Below the relationships between equation 9.8 and the four above mentioned transformations are listed.

	a_1	a_2	a_3	b_1	b_2	b_3
Translation	1	0	Δ_x	0	1	Δ_y
Scaling	S_x	0	0	0	S_y	0
Rotation	$\cos \theta$	$-\sin \theta$	0	$\sin \theta$	$\cos \theta$	0
Shearing	1	B_x	0	B_y	1	0

Often *homogeneous coordinates* are used when implementing the transformation since they make further calculations faster. In homogeneous coordinates, the affine transformation becomes:

projective coordinates $ah \sim h$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (9.9)$$

where $a_3 = \Delta x$ and $b_3 = \Delta y$.

"Introduction to Video and Image Processing",
Thomas B. Moeslund,
Springer, 2012, ISBN13:
9781447125020

Linear transformations (inverse)

If $\bar{A}$ is square ($n \times n$) and non-singular, $\bar{A}^{-1}$ is defined, thus

$$\bar{x} = \bar{F}^{-1}(\bar{y}) = \bar{A}^{-1} \bar{y}$$

An orthogonal ($n \times n$) matrix is one such, $\bar{A}^{-1} = \bar{A}^T$, i.e. a rotation in 2D and 3D Euclidean space — preserves the inner product (length)

Basis representations

Row and Column space of $\bar{A}$

Given basis for column space, $B = \text{Col}(\bar{A})$, any column vector can be expressed in B since $\bar{b}_i \in \text{Span}(B)$

Note

$\text{Col}(\bar{A}) \in \mathbb{R}^m$ and $\text{Row}(\bar{A}) \in \mathbb{R}^n$
are generally different subspaces.

Basis representation

If $B = \{\bar{b}_1, \dots, \bar{b}_k\}$ is a basis for vector space V , any $\bar{v} \in V$ can be written

$\bar{v} \in \text{Span}(B)$

$$\bar{v} = c_1 \bar{b}_1 + \dots + c_k \bar{b}_k$$

which we define as $[\bar{v}]_B = \bar{v}$
expressed on the basis B , with
coordinates $c_1, \dots, c_k$ wrt. B

Basis and linear transformations

If $\bar{v}$ has representation $[\bar{v}]_B$

$$\bar{v} = c_1 \bar{b}_1 + \dots + c_k \bar{b}_k$$

$$T(\bar{v}) = c_1 T(\bar{b}_1) + \dots + c_k T(\bar{b}_k) \quad \text{or,}$$

$$[T(\bar{v})]_C = c_1 [T(\bar{b}_1)]_C + \dots + c_k [T(\bar{b}_k)]_C$$

$$\text{Assume } M = \{[T(\bar{b}_1)]_C, \dots, [T(\bar{b}_k)]_C\}$$

$$[T(\bar{v})]_C = M [c_1 \dots c_k]^T = M [\bar{v}]_B$$

Orthogonal bases

If the basis constitutes an orthogonal set we have linear independence since $c_1 \bar{b}_1 + \dots + c_k \bar{b}_k = \bar{0}$

$$\Leftrightarrow \bar{b}_i \cdot (c_1 \bar{b}_1 + \dots + c_i \bar{b}_i + \dots + c_k \bar{b}_k) = \bar{0}$$

$$\Leftrightarrow c_i \bar{b}_i \cdot \bar{b}_i = c_i \|\bar{b}_i\|^2 = 0$$

$$\Leftrightarrow c_i = 0, i = 1 \dots k, \text{ so}$$

$$[\bar{v}]_{\mathcal{B}} \text{ has coordinates } c_i = \frac{\bar{v} \cdot \bar{b}_i}{\bar{b}_i \cdot \bar{b}_i}$$

Orthogonalization (Gram-Schmidt)

$$B = \{\bar{b}_1, \dots, \bar{b}_k\} \rightarrow B_\perp = \{\bar{v}_1, \dots, \bar{v}_k\}$$

$$\bar{v}_1 = \bar{b}_1$$

for $i = 2 : k$

$$\bar{v}_i = \bar{b}_i - \left(\frac{\bar{b}_i \cdot \bar{v}_1}{\bar{v}_1 \cdot \bar{v}_1} \bar{v}_1 + \dots + \frac{\bar{b}_i \cdot \bar{v}_{i-1}}{\bar{v}_{i-1} \cdot \bar{v}_{i-1}} \bar{v}_{i-1} \right)$$

end

⚠ Orthogonal $\rightarrow$ normalize $\frac{\bar{v}_i}{\|\bar{v}_i\|_2}$

Eigenvalue Concept

Eigenvalue problems

One special and important linear transformation is : $\bar{A}$ ($n \times n$)

$$\bar{A} \bar{x} = \lambda \bar{x}$$

where the image is a scaled version of the input !

This arises in many engineering problems, known as the matrix eigenvalue problem

Fibonacci numbers

$$f_{n+1} = f_n + f_{n-1}$$

$$P(n): f_{n+1}/f_n \approx \text{constant}, n \rightarrow \infty$$

$$\begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = \alpha \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix}, n \rightarrow \infty$$

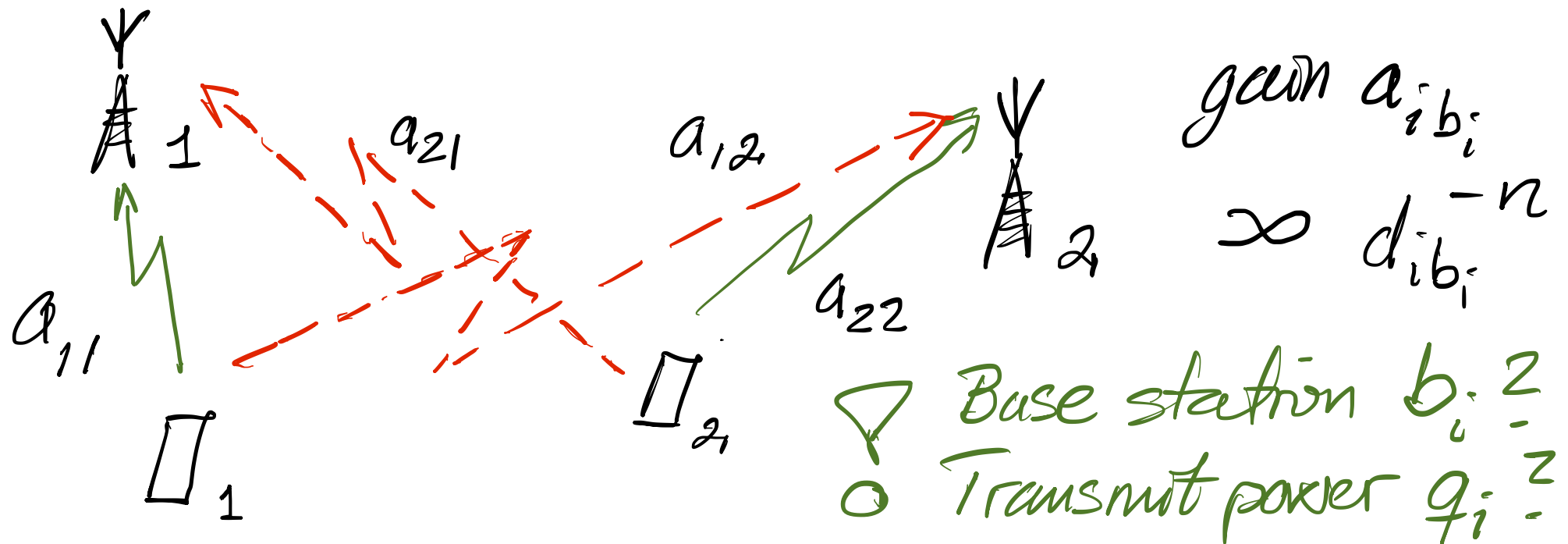
$$\begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} \approx \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix}$$

eigenvalue
problem

⇓

$$\alpha \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix} \quad \alpha \bar{F} = \bar{A} \bar{F}$$

Power control in wireless



In reception at base station b_i from mobile i the signal to interference is

$$\omega_i = q_i a_{ib_i} / q_j a_{jb_i}$$

Power control in wireless (cont.)

eg. mobile 1 connected to $b_1 = 1$ and
mobile 2 connected to $b_2 = 2$

$$w_1 = \frac{q_1 \cdot a_{11}}{q_2 \cdot a_{21}}$$

Would like to max w_i (max rate)
and enforce $w_i = w$ (same service)

eigenvalue
problem

$$\frac{1}{w} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 0 & a_{2b_1}/a_{1b_1} \\ a_{1b_2}/a_{2b_2} & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$$